

Certificate Manager Enrollment Over Secure Transport

cerebus2

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1 About

`estissuer` is a cert-manager External certificate issuer that implements a subset of RFC 7030 Enrollment over Secure Transport (EST). It provides custom resources for EST services and a custom controller that reconciles certificate requests.

`est-operator` is written in Python using the kopf framework and lightkube.

This operator covers only a portion of RFC 7030; see this document for an overview of what's implemented, planned, and not planned.

See this file for an overview of this operator's design.

2 Getting Started

2.1 Prerequisites

2.2 Building

2.3 Installing

See 4, below. Also see the design document for a deployment description.

3 Testing

3.1 Style

3.2 Component

3.3 End-to-end

4 Deployment

4.1 In cluster

TBD.

4.2 Out of cluster

5 Usage

- EstIssuer and EstClusterIssuer secrets must be passwords and labeled "est.mitre.org/registration=true".
- Certificate secrets must be labeled "est.mitre.org/certificate=true".
 - RFC 7030 renewals require mTLS authentication even when initial renewal uses a password registration credential, so this label allows the controller to track certificate private keys.

6 Federal Information Processing Standards (FIPS) 140-2 certified mode

This controller can be built to satisfy FIPS 140-2 certification requirements. FIPS 140-2 certification depends on the module and the platform. A platform with Python dynamically linked to OpenSSL *and* links OpenSSL to a validated version of the OpenSSL FIPS Object Module can demonstrate FIPS 140-2 certification. Consult the NIST Cryptographic Module Validation Program list of OpenSSL modules for combinations of OpenSSL and platform that are certified.

Note: OpenSSL 1.0.2 is no longer supported by the OpenSSL project, but OpenSSL FIPS Object Module 2.0 is not compatible with OpenSSL 1.1. A new version is not available nor certified yet.

Other FIPS certified modules are available (e.g., GnuTLS and Network Security Services) but the work to integrate these modules is extensive. We welcome merge requests on this topic.

6.1 Building a FIPS version

6.2 Building a FIPS container image

6.3 Running a FIPS version out of cluster

7 Built with

8 Contributing

TBD

9 Versioning

TBD

10 Authors

- Timothy J. Miller - *initial work* - Cerebus

11 License

Apache License v2.0

12 Acknowledgements

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- Sergey Vasilyev for kopf, so I don't have to learn Go yet.